

REPMS: Real Estate Property Management System

Mini Project Description & Technical Specification | Multi-Asset Web UI Architecture

Project Title:	REPMS (Real Estate Property Management System)	Academic Section:	Section AB
Domain / Track:	Enterprise Web UI Engineering / PropTech Systems	Project Group No:	Group 15
Core Architecture:	4-Developer Conflict-Free Modular BEM CSS & Token System	File Designation:	REPMS_AB_15_desc.pdf
Technology Stack:	HTML5, CSS3 (Design Tokens, Grid, Flexbox, BEM), Extensible to JS/React	Team Members:	Shourya (Dev 1), Shikhar (Dev 2), Shatakshi (Dev 3), Shakti Dubey (Dev 4)

1. Executive Summary

The Real Estate Property Management System (REPMS) is an enterprise-grade, high-density web application designed to streamline operations across all major real estate asset classes—including Residential, Commercial Office, Retail Malls, Industrial Logistics, Mixed-Use complexes, and Vacant Land. Commercial real estate enterprises manage multimillion-dollar portfolios using outdated or disconnected tools, resulting in delayed financial audits and inefficient tenant-vendor communication. REPMS bridges this operational gap by delivering a unified, production-ready web UI prototype. Built strictly on standards-compliant HTML5 and CSS3 (leveraging Design Tokens, BEM naming, and pure CSS interactivity), the system enables four developers—Shourya, Shikhar, Shatakshi, and Shakti Dubey—to build concurrently with zero stylesheet collisions, creating a scalable foundation for future JavaScript event hydration and React component migration.

2. Problem Statement & Industrial Need

Real estate asset managers face critical pain points: (a) Fragmented Tools: Residential and commercial assets are managed on disparate software, preventing unified Net Operating Income (NOI) analytics; (b) Delayed Financial Audits: Manual rent rolls and unlinked CAM reconciliations introduce accounting delays; (c) Unstructured Maintenance: Emergency work orders lack clear priority dispatching, causing tenant dissatisfaction; and (d) Frontend Collaboration Friction: Concurrent multi-developer styling frequently causes catastrophic CSS cascade collisions. REPMS systematically addresses these challenges through domain-driven modular UI design and strict scoped CSS architecture.

3. Mini Project Objectives

The core objectives of the REPMS project are formulated as follows (5–10 specific operational goals):

- 1. Centralized Multi-Asset Management: Build a unified platform interface tracking residential, commercial, retail, industrial, mixed-use, and vacant land properties.
- 2. Real-Time Executive Portfolio Dashboards: Provide asset managers with instantaneous visibility into total portfolio value, blended occupancy rates, and critical alert feeds.
- 3. Unit-by-Unit Stacking Visualizer: Design interactive building floorplan visualizers and floor-by-floor unit stacks with color-coded vacancy and occupancy states.
- 4. Complete Tenant & Lease Lifecycle Tracking: Streamline tenant directories, lease term tracking, automated rent escalation schedules, and security deposit escrow monitoring.
- 5. Financial Operations & Rent Roll Auditing: Model Net Operating Income (NOI) calculation views, monthly rent roll statements, payment ledgers, and interactive invoice generation.
- 6. Prioritized Maintenance Dispatch & SLA Kanban: Implement a priority-based work order pipeline (Emergency, High, Medium, Low) with vendor assignment cards and SLA countdowns.
- 7. Leasing CRM & Prospect Funnel: Build public/internal vacancy listing galleries, prospective tenant inquiry stages (Tour Scheduled, Applied, Approved), and booking flows.
- 8. Sustainability (ESG) & Yield Analytics: Integrate property yield comparison charts, energy efficiency scorecards, and carbon offset gauges to support environmental compliance.
- 9. Zero-Collision 4-Developer CSS Architecture: Engineer a structured 5-tier CSS token hierarchy (Primitives, Semantics, Components, Scoped Page Namespaces) ensuring seamless team collaboration.

4. System Scope & Multi-Asset Coverage

REPMS is architected to address the operational and financial requirements of six distinct real estate asset categories:

Asset Category	Target Typology & Facility Operations	Key Tracked Operational Metrics
Residential & Multi-Family	Apartments, Condos, Gated Communities, Single-Family Units	Occupancy rate, tenant turnover, monthly rent collection, move-in logs
Commercial Office Towers	Class-A Towers, Co-Working Floors, Corporate Campuses	Price per sq.ft, lease escalation schedules, multi-year CAM charges
Retail Malls & Centers	Shopping Centers, Anchor Outlets, Retail Kiosks, Strip Malls	Footfall index, base rent vs. revenue share terms, anchor lease expiries
Industrial Logistics Parks	Warehouses, Distribution Centers, Cold Storage Facilities	Clear ceiling height, dock door capacity, heavy electrical specs, logistics access
Mixed-Use Complexes	Integrated Residential + Retail + Commercial Developments	Zoning allocations, shared facility maintenance, blended portfolio yield
Vacant Land & Plots	Commercial Zoned Parcels, Development Sites, Ground Leases	Total acreage, municipal zoning status, land lease holding yield

5. Key Functional Modules & System Specification

The application features eight dedicated operational modules distributed across separate screen templates:

Module & Target Page	Functional Capabilities & User Workflows	UI Components Used	Module Lead
Executive Master Dashboard (index.html)	Real-time portfolio overview, cross-asset KPI summary cards (Total Assets, Portfolio Value, Avg. Occupancy, Monthly Revenue), urgent maintenance feed, and occupancy visualizer.	Metric Cards, Stat Badges, Notification Banners, Quick-Action Toolbars	Shourya (Dev 1)
Portfolio & Unit Stacker (pages/portfolio.html)	Property catalog grouped by asset class, interactive building floorplan visualizer, floor-by-floor unit stack view with vacancy color indicators.	Asset Cards, Unit Stack Grid, Floor Selector Tabs, Spec Badges	Shourya (Dev 1)
Tenant Directory (pages/tenants.html)	Multi-tenant registry, tenant profile modal views, emergency contact directories, active lease status pills, unit linkages, and payment ledger history.	Data Tables, Search/Filter Toolbar, Detail Modal, Status Badges	Shikhar (Dev 2)
Lease Lifecycle Hub (pages/leases.html)	Master repository of active, pending, and expiring leases; rent escalation schedules; security deposit escrow accounts; lease clause checklists.	Status Badges, Filter Tabs, Escalation Timeline, Expiry Alerts	Shikhar (Dev 2)
Financials & Rent Roll (pages/financials.html)	Monthly Rent Roll statements, Net Operating Income (NOI) tracker, revenue vs. expense breakdown, overdue payment alerts, pure CSS invoice generator modal.	Financial Summary Cards, High-Density Data Tables, Pure CSS Invoice Modal	Shatakshi (Dev 3)
Maintenance & Dispatch (pages/maintenance.html)	Priority-based Work Order pipeline (Emergency, High, Medium, Low), SLA countdown visualizers, vendor assignment cards, maintenance expense ledger.	Work Order Kanban Grid, SLA Badges, Priority Indicators, Dispatch Drawer	Shatakshi (Dev 3)
Leasing CRM & Funnel (pages/leasing.html)	Public and broker-facing vacancy showcase, applicant pipeline stages (Inquiry, Tour Scheduled, Application Submitted, Approved), tour scheduling UI.	Listing Cards, Pipeline Kanban, Tour Schedule Picker, Application Form	Shakti Dubey (Dev 4)
Analytics & ESG Reports (pages/analytics.html)	Portfolio yield comparison charts, energy efficiency rating scorecards, water & electricity consumption trackers, carbon offset indices.	Pure CSS Bar Charts, Donut Meters, ESG Gauges, Analytical Metric Tables	Shakti Dubey (Dev 4)

6. Technical Architecture & 4-Developer CSS Methodology

To enable concurrent engineering by four developers without styling collisions or git merge friction, REPMS implements a 5-tier layered stylesheet architecture utilizing CSS Custom Properties and strict BEM conventions:

- Tier 1: Primitives (tokens/primitives.css): Raw foundational variables—hex colors, 8pt spacing grid, typography stacks (Inter, Outfit), and elevation box-shadows.
- Tier 2: Semantic Intent Tokens (tokens/semantic.css): Functional mappings such as --color-brand-primary, --color-surface-card, and --color-badge-urgent.
- Tier 3: Base & Layout Engine (base/reset.css, base/layout.css): CSS normalization, responsive grid app shells, collapsible sidebars, and top navigation header bars.
- Tier 4: Shared Reusable BEM Components (components/*.css): Encapsulated UI blocks (buttons, data tables, cards, badges, progress meters, modals) styled strictly without styling raw HTML tags.
- Tier 5: Scoped Page Namespaces (pages/*.css): Strict root container scoping (e.g., .page-financials, .page-maintenance) guaranteeing zero cascade cross-contamination.
- Pure CSS Interactivity: Tab switches, filter toggles, slide-out drawer panels, and popup modals function via pure CSS pseudo-states (:checked, :focus-within, :hover).

7. Team Task Distribution & Work Breakdown Structure (WBS)

Responsibilities and deliverables are partitioned across team members according to modular ownership:

Team Member & Role	Module Ownership & Functional Responsibilities	Target Code Deliverables
Shourya Lead Architect & Shell Lead (Dev 1)	Design token foundation, master App Shell grid (Sidebar, Header, Main, Footer), Executive Dashboard UI, and Property/Unit Stacker directory.	base/reset.css, base/layout.css index.html, pages/portfolio.html
Shikhar Tenant & Lease Lead (Dev 2)	High-density data tables, status badges, filter toolbars, Tenant Directory with profile drawers, and Master Lease lifecycle management page.	components/tables.css, badges.css pages/tenants.html, pages/leases.html
Shatakshi Financials & Maintenance Lead (Dev 3)	Financial metric cards, Rent Roll statements, Pure CSS invoice generator modal, Work Order Kanban pipeline, and Vendor dispatch SLA trackers.	components/cards.css, modals.css pages/financials.html, pages/maintenance.html
Shakti Dubey Leasing CRM & Analytics Lead (Dev 4)	Vacancy showcase gallery, prospective lead CRM pipeline, tour scheduling interface, portfolio yield charts, and ESG sustainability scorecards.	components/progress.css, charts.css pages/leasing.html, pages/analytics.html

8. Project Deliverables, Expected Outcomes & Future Roadmap

- Comprehensive 8-Screen UI Prototype: Fully responsive, cross-browser compatible application covering all aspects of property management.
- Zero External Dependencies: High-performance pure HTML5/CSS3 architecture without heavy client-side JavaScript or CSS framework overhead.
- Extensible Codebase: Semantic DOM structures and BEM class names architected for immediate JavaScript hydration and future React component refactoring.
- Enterprise Evaluation Readiness: Realistic multi-asset sample datasets demonstrating enterprise viability for real-world real estate operations.

Project Title: REPMS	Section & Group: Section AB — Group 15	Submission Format: PDF & Text Objectives
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